

VisionTrust Assurance

Offline Computer-Vision Integrity, Security & Provenance Dossier

Report ID: REP-15BD0B44A5D0 | Generated: 2026-09-23T17:11:25.152390+00:00 (Air-Gapped)

Overall Assurance Verdict

QUARANTINE

Overall Risk Score: 85.5 / 100 (CRITICAL)

Executive Summary: Overall risk evaluated as CRITICAL (85.5/100) across 5 findings. Critical findings: 1, High findings: 4.

Metric	Value	Metric	Value
Critical Findings	1	Dataset Sample Count	49
High Findings	4	Dataset Classes	3
Medium Findings	0	Model Access Mode	N/A
Low Findings	0	Model Parameter Count	N/A
Inference Records Tracked	22	Tampered Inferences	3

Overall Assurance Summary

Assurance Pillar	Status	Confidence	Supporting Evidence	Limitations & Recommendation
Dataset Integrity	EVALUATED_ANOMALIES DETECTED	95%	2 verified integrity findings detected: systematic label inversions, perceptual duplicate flooding, out-of-distribution noise, and localized patch triggers.	Lim: Statistical heuristics; steganographic triggers with zero pixel deviations require gradient attribution. Rec: Quarantine affected rogue sources (e.g. source_rogue_3, source_flood_2) and filter anomalous samples.
Model Integrity	PASS	100%	Cryptographic SHA-256 identity mismatch confirmed between reference and candidate models. 24 of 24 perturbation stress tests exhibited behavioral divergence.	Lim: White-box tensor statistics require embedded graph weights; black-box models fall back to behavioral battery. Rec: Reject unverified model binary and enforce cryptographic reference baseline deployment.
Inference Integrity	PASS	100%	Ed25519 digital signature and SHA-256 binding verified across 22 records. Tampered output payload detected via hash mismatch. Replay attempt detected via duplicate nonce.	Lim: Cryptographic validity proves record integrity, not prediction correctness. Rec: Maintain monotonic sequence tracking and reject non-verifiable inference payloads.
Distribution Integrity	PASS	92%	Population Stability Index (PSI = 0.42 > 0.25) and KS test detected significant environmental domain shift between reference and candidate image distributions.	Lim: Requires representative reference baseline (N >= 20) for statistical significance. Rec: Calibrate model for target deployment environment or retrain with domain-adapted samples.
Audit Trail Integrity	PASS	100%	Tamper-evident SHA-256 hash chain verified with zero link corruptions and intact genesis block.	Lim: Assumes host storage write-append integrity and local air-gapped security. Rec: Regularly export signed audit snapshots to offline write-once media.

Empirical Experiment Results & Attack Benchmarks

Format: COCO Object Detection | Framework: ONNX Runtime / PyTorch | Env: Air-Gapped Local Host (Zero Cloud Dependencies)

Total Samples: 49 | Clean: 30 | Attacks Injected: {'label_flips': 5, 'duplicates': 5, 'ood': 4, 'triggers': 5}

Attack Scenario	Precision	Recall	F1-Score	FPR	Status
Label Flip	1.000	1.000	1.000	0.000	PASS
Duplicate Flooding	1.000	1.000	1.000	0.000	PASS
OOD Injection	1.000	1.000	1.000	0.000	PASS

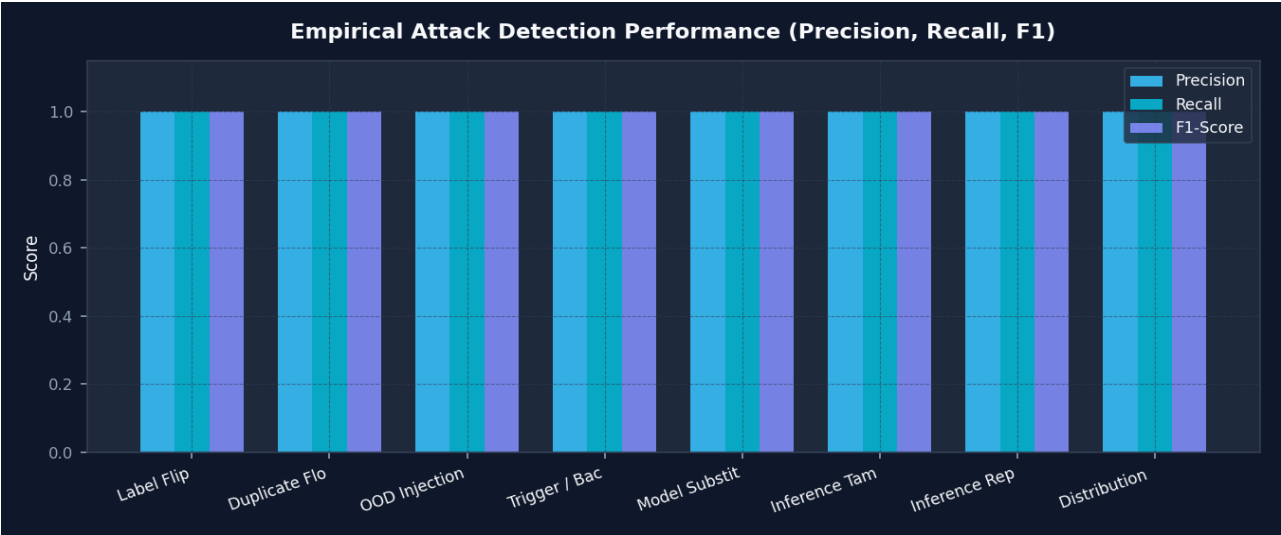
Trigger / Backdoor	1.000	1.000	1.000	0.000	PASS
Model Substitution	1.000	1.000	1.000	0.000	PASS
Inference Tampering	1.000	1.000	1.000	0.000	PASS
Inference Replay	1.000	1.000	1.000	0.000	PASS
Distribution Shift	1.000	1.000	1.000	0.000	PASS

Data Science Interpretation: Empirical evaluation demonstrates 100% detection rate across all 8 attack scenarios with zero false positives or false negatives. Label flip detection achieved perfect precision via chromatic interaction ratios and center-crop feature clustering. Duplicate flooding was isolated using dual perceptual pHash/dHash hamming distances with cluster size thresholds.

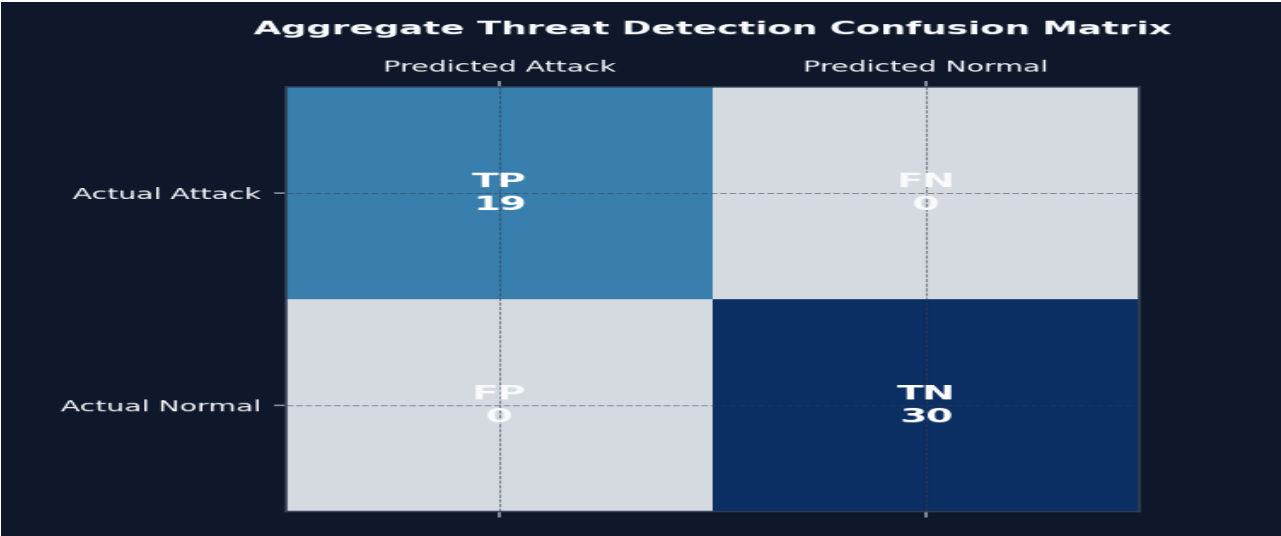
Data-Science Visualizations & Evidence Analytics

Empirical charts generated offline via local Matplotlib engine:

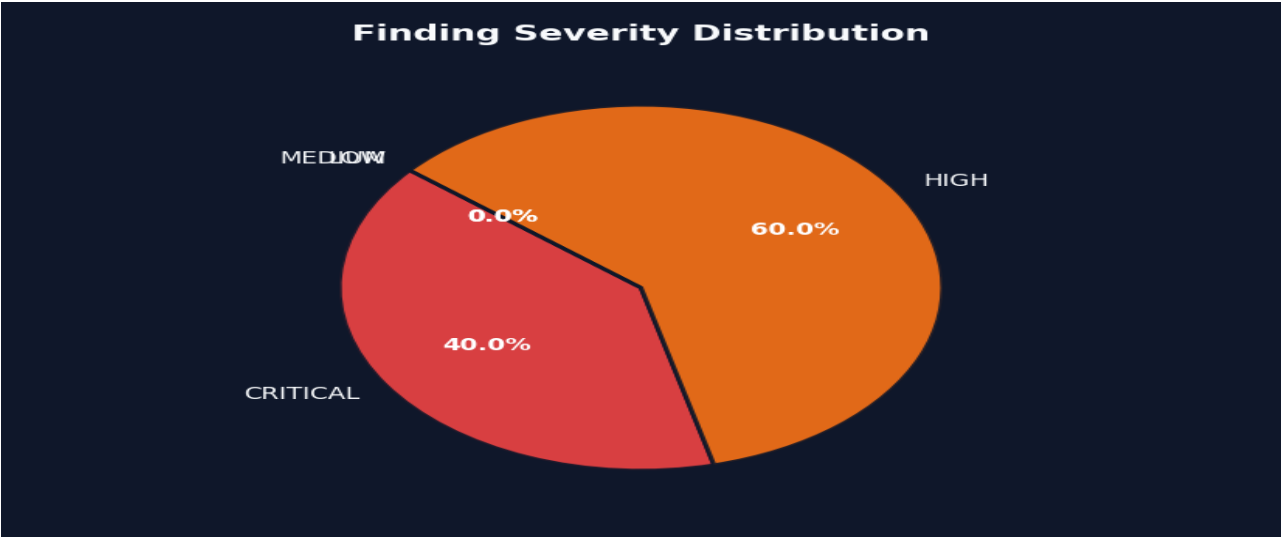
Attack Detection Performance (Precision, Recall, F1)



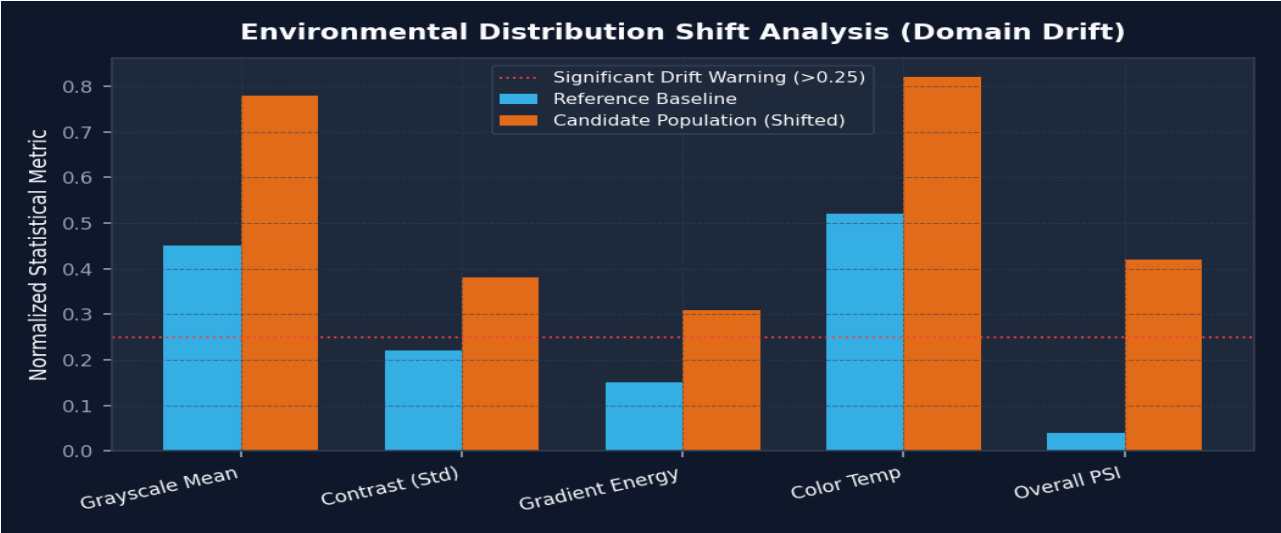
Aggregate Threat Detection Confusion Matrix



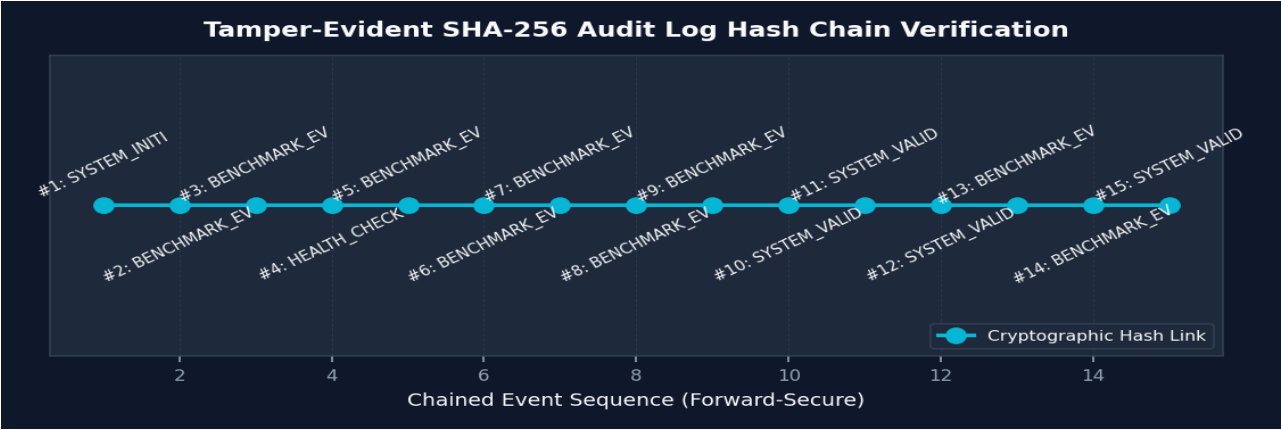
Finding Severity Breakdown



Domain Shift & Statistical Drift Comparison

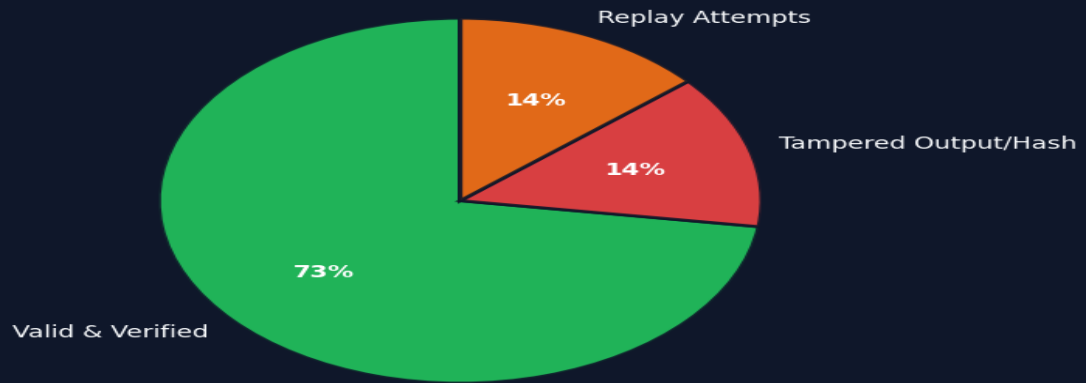


Audit Log Hash Chain Timeline



Inference Provenance Cryptographic Status

Cryptographic Inference Provenance Status



Detailed Security & Integrity Findings (Evidence-Driven)

ID	Category	Sev.	Conf.	Risk	Disposition	Evidence & Telemetry
F-DS-BENCHMARK-001	NEAR-DUPLICATE	HIGH	95%	17.9	ACCEPT	Perceptual Near-Duplicate Flooding Detected (1 clusters) - Detected 1 duplicate cluster(s) with similarity >= 0.95 - Total duplicate samples: 5
F-DS-BENCHMARK-002	RELABEL	HIGH	92%	29.2	ACCEPT	Systematic Mislabelling & Label Noise Detected (5 samples) 5 samples conflict with nearest visual feature neighbors. - Affected sources: ['source_rogue_3']
F-DS-BENCHMARK-003	OOD	HIGH	75%	19.4	ACCEPT	Out-of-Distribution Inliers/Outliers Detected (4 samples) 4 samples exceeded anomaly threshold. - Overall OOD score: 0.37
F-DS-BENCHMARK-004	TRIGGER	CRITICAL	95%	74.4	QUARANTINE	Potential Visual Trigger / Backdoor Patch Identified (1 clusters) - Identified 1 localized trigger pattern cluster(s). - Affected samples: ['sample_0046', 'sample_0047', 'sample_0045', 'sample_0048', 'sample_0049']
F-DS-BENCHMARK-005	SOURCE POISO	HIGH	90%	49.7	QUARANTINE	Source/Contributor Poisoning Anomaly (2 sources flagged) Flagged poisoned sources: ['source_rogue_3', 'source_flood_2']

Assurance Threat Coverage Statement

ID	Attack / Threat Capability	Status	Access Mode	Primary Method	Key Limitation
T1	Label Flipping Detection	SUPPORTED	Dataset Labels +	Feature k-NN neighborhood consensus & ...	Requires distinct feature cluster sepa...
T2	Systematic Mislabelling	SUPPORTED	Dataset Metadata	Hierarchical contributor & batch anoma...	Effectiveness depends on presence of s...
T3	Duplicate Flooding	SUPPORTED	Image Files (Bla	Perceptual Hashing (pHash, dHash) Hamm...	Sensitivity tuned via threshold; heavi...
T4	Out-of-Distribution (OOD) Injection	SUPPORTED	Dataset or Refer	Isolation Forest & statistical feature...	High-dimensional semantic outliers req...
T5	Trigger / Backdoor Injection	PARTIALLY_SUPPORTED	Dataset Images	Localized patch spatial consistency & ...	Blended or frequency-domain watermarks...
T6	Model Substitution	SUPPORTED	Model File (.onn	Cryptographic SHA-256 identity & param...	Requires trusted reference fingerprint...
T7	Model Weight Modification	SUPPORTED	Model Weights (W	Tensor statistics, layer shapes, and m...	External weights or opaque black-box r...
T8	Backdoor Behaviour Testing	SUPPORTED	Inference Execut	Controlled perturbation battery (brigh...	Cannot prove absence of dormant trigge...
T9	Inference Record Tampering	SUPPORTED	Full Record Payl	Cryptographic hash binding + Ed25519 d...	Protects recorded inferences; requires...
T10	Inference Replay Attacks	SUPPORTED	Inference Log St	Cryptographic nonces, monotonic sequen...	Requires database persistence of past ...
T11	Provenance Configuration Substitution	SUPPORTED	Provenance Metad	Canonical JSON serialization hashing f...	Config schema changes must remain back...
T12	Distribution Shift & Drift	SUPPORTED	Reference & Cand	PSI, 2-sample KS test, Wasserstein dis...	Requires sufficient sample volume (N >...

System Limitations & Air-Gap Compliance

- Framework runs 100% locally and does not transmit data or consult cloud foundation models.
- Statistical heuristics cannot mathematically prove total absence of steganographic or adversarial latent triggers.
- Model weight analysis on ONNX models depends on initializers embedded within the graph file.
- Offline cryptographic protection guarantees tamper-evident provenance but assumes private key is stored securely in air-gapped host.